LAB REPORT 2
K1 CSE3023 PEMBANGUNAN APLIKASI BERASASKAN WEB
SEM 4

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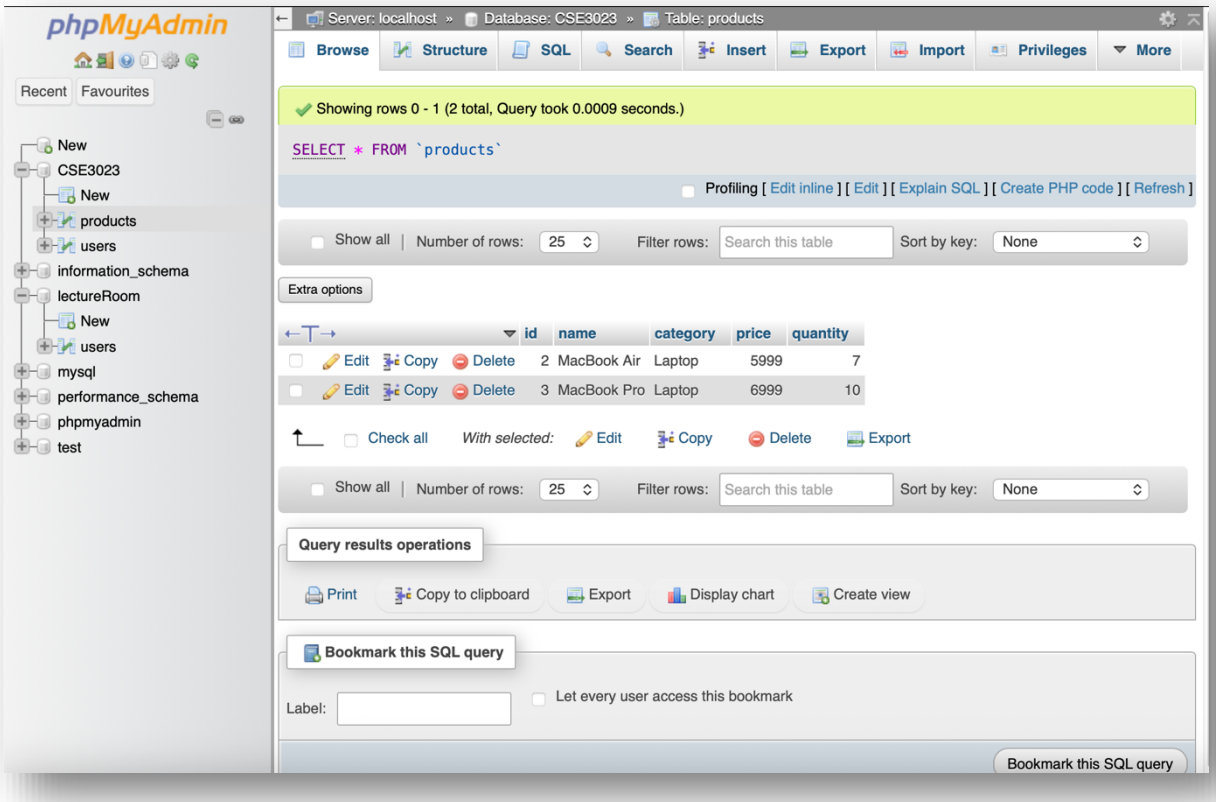
Exercise

Database



Source Code

DeleteServlet.java

```java
 */
package com.lab.controller;
import com.lab.dao.ProductDAO;
import java.io.IOException;
import java.io.PrintWriter;
//import java.sql.Connection;
//import java.sql.DriverManager;
//import java.sql.ResultSet;
//import java.sql.Statement;
//import java.sql.Connection;
//import java.sql.DriverManager;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

/**
 *
 * @author sazliosman
 */
@WebServlet("/DeleteServlet")
public class DeleteServlet extends HttpServlet {

    private ProductDAO productDAO;
    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */

    public void init() {
        productDAO = new ProductDAO();
    }
```

```java
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet DeleteServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet DeleteServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the c
    /**
     * Handles the HTTP <code>GET</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String idStr = request.getParameter("id");

        if (idStr !=null && !idStr.isEmpty()) {
            try {
                int id = Integer.parseInt(request.getParameter("id"));
                productDAO.deleteProduct(id); //Panggil DAO
            }catch (NumberFormatException e) {
                e.printStackTrace();
```

Inserservlet.java

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
//import java.sql.Connection;
//import java.sql.DriverManager;
//import java.sql.PreparedStatement;

/**
 *
 * @author sazliosman
 */
@WebServlet("/InsertServlet")
public class InsertServlet extends HttpServlet {

    private ProductDAO productDAO;

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */

    public void init() {
        productDAO = new ProductDAO();
    }
```

```java
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        //1. capture user inputs from HTML form
        String name = request.getParameter("name");
        String category = request.getParameter("category");
        Double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        //2. create product object
        Product newProduct = new Product(0, name, category, price, quantity);

        //3. call DAO method
        productDAO.insertProducts(newProduct); //Panggil DAO, bukan JDBC
        response.sendRedirect("ViewServlet");
;    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

Viewservlet.java

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
//import java.sql.DriverManager;
//import java.sql.ResultSet;
//import java.sql.Statement;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.util.List;

/**
 *
 * @author sazliosman
 */
@WebServlet("/ViewServlet")
public class ViewServlet extends HttpServlet {

    private ProductDAO productDAO;

```

```java
response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        //fetch data from DAO
        List<Product>listProduct = productDAO.selectAllProduct();

        out.println("<html><body>");

        out.println("<table border='1'>");
        out.println("<h2>User List(Using DAO</h2>");
        out.println("<table border='1'<tr><th>ID</th><th>Name</th><th>Category </th><th>Price</th><th>Quantity</th><t

        for (Product product : listProduct) {
            out.println("<tr>");
            out.println("<td>" + product.getId() + "</td>");
            out.println("<td>" + product.getName() + "</td>");
            out.println("<td>" + product.getCategory() + "</td>");
            out.println("<td>" + product.getPrice() + "</td>");
            out.println("<td>" + product.getQuantity() + "</td>");


            out.println("<td><a href='UpdateServlet?id=" + product.getId() + "'>Edit</a> |");
            out.println("<a href='DeleteServlet?id=" + product.getId() + "'> Delete</a></td>");
            out.println("</tr>");
        }
        out.println("</table>");
        out.println("<br><a href='index.html'>Add New User</a>");

        }
```

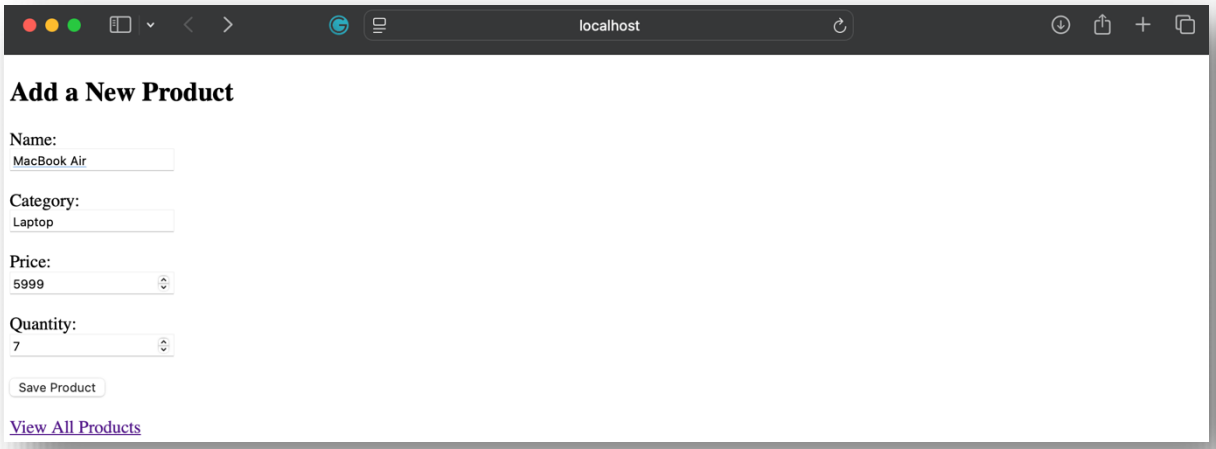
Updateservlet.java

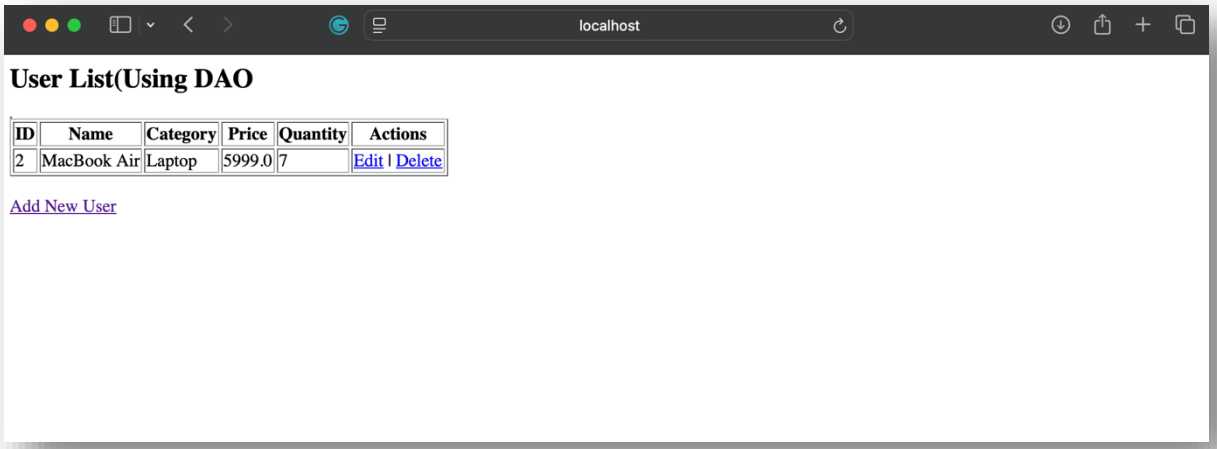
```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;

import java.io.IOException;
import java.io.PrintWriter;
//import java.sql.Connection;
//import java.sql.DriverManager;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

/**
 *
 * @author sazliosman
 */

@WebServlet("/UpdateServlet")
public class UpdateServlet extends HttpServlet {
```

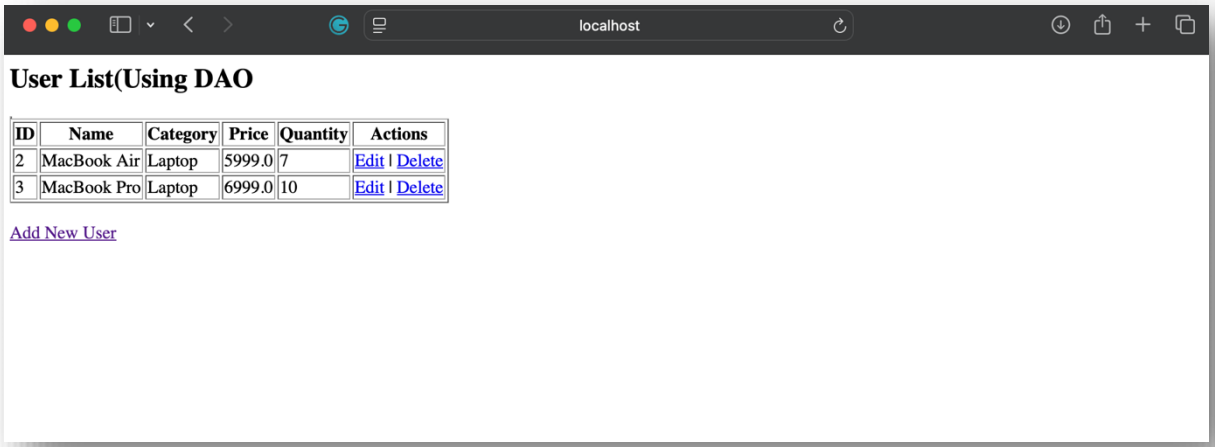
```java
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    int id = Integer.parseInt(request.getParameter("id"));
    Product existingProduct = productDAO.selectProduct(id); //Panggil DAO
    if(existingProduct !=null) {
        out.println("<h2>Update Product (Using DAO)</h2>");
        out.println("<form action='UpdateServlet' method='POST'>");
        out.println("<input type='hidden' name='id' value='" + existingProduct.getId() + "'>");
        out.println("Name: <input type='text' name='name' value='" + existingProduct.getName() + "' required><br><br>");

        out.println("Category: <input type='text' name='category' value ='" + existingProduct.getCategory() + "' required><br><br>");
        out.println("Price: <input type='number' name='price' value='" +existingProduct.getPrice() + "' required><br><br>");
        out.println("Quantity: <input type='number' name='quantity' value='" +existingProduct.getQuantity() + "' required><br><br>");

        out.println("<input type='submit' value='Update Product'>");
        out.println("</form");
    }
}
```

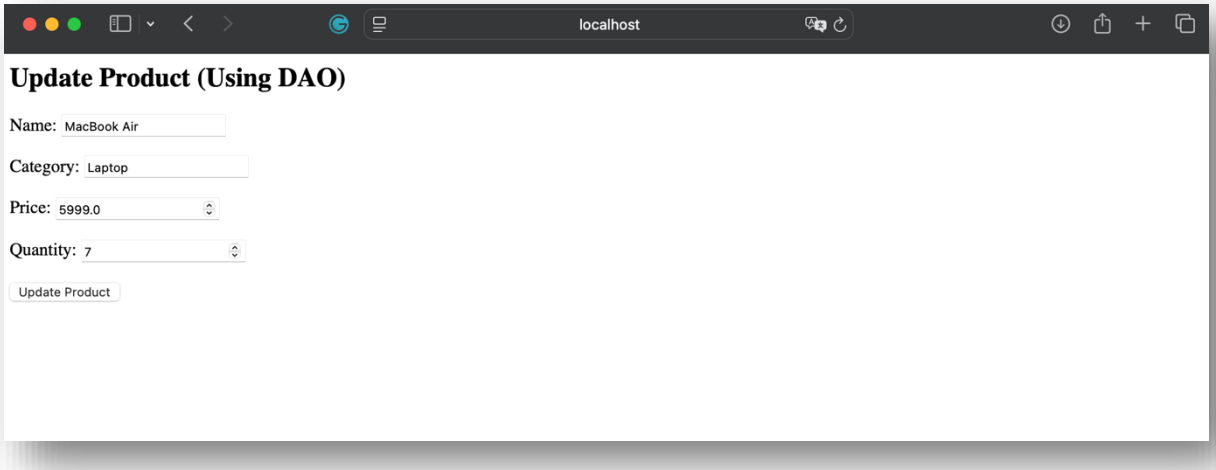
```java
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    int id = Integer.parseInt(request.getParameter("id"));
    String name = request.getParameter("name");
    String category = request.getParameter("category");
    Double price = Double.parseDouble(request.getParameter("price"));
    int quantity = Integer.parseInt(request.getParameter("quantity"));

    Product product = new Product(id, name, category, price, quantity);
    productDAO.updateProducts(product);

    response.sendRedirect("ViewServlet");
}
```

Output


localhost
Add a New Product
Name:
MacBook Air
Category:
Laptop
Price:
5999
Quantity:
7
Save Product
View All Products


localhost
User List(Using DAO
ID
Name
Category
Price
Quantity
Actions
2
MacBook Air
Laptop
5999.0
7
Edit | Delete
Add New User


localhost
User List(Using DAO
ID | Name | Category | Price | Quantity | Actions
2 | MacBook Air | Laptop | 5999.0 | 7 | Edit | Delete
3 | MacBook Pro | Laptop | 6999.0 | 10 | Edit | Delete
Add New User


localhost
Update Product (Using DAO)
Name: MacBook Air
Category: Laptop
Price: 5999.0
Quantity: 7
Update Product